

Project Executable Files

Date	07 July 2026
Team ID	SWTID-2026-8075
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	No
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	No

9	Test files and test results	No
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

/Voice-Based-Concept-Understanding-Analyser

```

|— main.py
|— app.py
|— requirements.txt
|— README.md
|— pdf_reportr.py
|— similarity.py

```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Deployed
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Localhost
Repository Link	https://github.com/Prayagbhardwaj03/Voice-Based-Concept-Understanding-Analyser

Demo Video Link	https://drive.google.com/file/d/1mUyQfj4JRQwzxLMbKa8LZrDNifbOTBWh/view
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Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run project locally:

Pre-requisites

Python 3.10 or above

Git

Visual Studio Code

Internet Connection

Required Python Libraries:

- Streamlit
- OpenAI Whisper
- Sentence Transformers
- Librosa
- SoundFile
- Matplotlib
- ReportLab

Steps

Step 1:

Download or clone the project repository.

Step 2:

Open the project folder in Visual Studio Code.

Step 3:

Install the required libraries.

```
pip install -r requirements.txt
```

Step 4:

Run the Streamlit application.

```
streamlit run app.py
```

Step 5:

Upload or record an audio file.

Step 6:

The application converts speech into text.

Step 7:

The system analyzes concept understanding and semantic similarity.

Step 8:

View the generated report and similarity score.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Large audio files require longer processing time	Optimize audio length before uploading
2	Background noise may reduce transcription accuracy	Use a clear audio recording
3	Internet may be required during initial model download	Download models before demonstration
4	Performance depends on system hardware	Recommended to use at least 8 GB RAM
5	Currently supports English language only (<i>if applicable</i>)	Future enhancement